



Toward Sustainable Banking Practices: Risk Management Committees, Environmental, Social, and Governance Practices, and Biodiversity Disclosure

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Abstract: The study aims to explore the role of the characteristics of the risk management committee (RMC) on biodiversity disclosure (BioDD), while analyzing the moderating role of Environmental, Social, and Governance (ESG) disclosure practices in the Iraqi banking sector. The importance of the study stems from the growing global interest in environmental disclosure and the role of corporate governance in improving transparency and sustainability in emerging economies. The study is based on a sample of 30 banks listed on the Iraqi Stock Exchange for the period from 2022 to 2025, with a total of 120 observations. Quantitative measurement of variables and multiple hierarchical regression analysis were used to test the hypotheses of the study. The results show that RMC characteristics have a positive and statistically significant association with BioDD, and ESG disclosure is significantly associated with the moderating effect observed in the RMC–BioDD relationship. In addition, bank size is positively related to BioDD. The study contributes to the emerging literature on BioDD by integrating RMC and ESG frameworks within agency, legitimacy, and stakeholder theories. It also offers practical implications for regulators and banking executives on integrating sustainability considerations into ESG risk frameworks to improve BioDD quality. Ultimately, this study addresses a literature gap regarding emerging markets and provides empirical evidence to inform sustainable banking practices in the Iraqi context.

Keywords: Sustainability challenges; Risk management committee; Biodiversity disclosure; Environmental, Social, and Governance; Sustainability reports

1. Introduction

Over the past two decades, the world has witnessed a radical shift in the nature of reports issued by institutions, as the evaluation of companies is no longer limited to traditional financial indicators only, but has extended to include the extent of their commitment to environmental and social responsibilities and corporate governance as indicators that show the ability of companies to achieve sustainability and maximize their value and reputation in the long term (Musa et al., 2025). This shift is due to escalating global environmental challenges, such as climate change, biodiversity loss, and the depletion of natural resources, as well as growing pressure from investors, regulators, and the international community demanding more transparent disclosures about the environmental and social impacts of the activities of these institutions (Kopnina et al., 2024). In this context, biodiversity has become one of the most complex modern sustainability issues, given its close and direct connection to the conservation of ecosystems and the sustainability of natural resources, and the wide-ranging economic, financial and operational impacts of their degradation.

Empirical evidence demonstrates that approximately 322 international financial institutions, including major banks and investment companies, have already turned to nature-related disclosure in accordance with TNFD

requirements (Goldberg & Yonge, 2024). Financial institutions have become more vulnerable to risks associated with financing companies that negatively affect biodiversity, especially with regard to the mining and oil sector, as it is linked to environmental pollution, water, and intensive agriculture, which helps banks identify future credit and environmental risks and enhance financial stability in the long term by avoiding investment in environmentally suspicious projects (Mundaca & Heintze, 2024). The importance of biodiversity disclosure (BioDD) in the financial sector is exemplified by HSBC, which has integrated biodiversity-related disclosure into its annual sustainability reports and adopts policies restricting the financing of activities that lead to deforestation. BNP Paribas has also imposed financing restrictions on certain industries with high environmental impact. While the Bank of China focused on green financing in line with the goals of sustainable development. In another context, Mundaca & Heintze (2024) confirmed that European banks are still relatively slow in measuring, evaluating, and disclosing the risks of ecosystem services. The study also indicated that approximately 26 cents per dollar of equity at these banks may be exposed to risks associated with ecosystem services.

The significance of BioDD lies in its role of enhancing transparency and limiting information asymmetry between management and stakeholders. The Taskforce on Nature-related Financial Disclosures (<https://tnfd.global/>), which initiative launched in 2021, found that risks related to nature and biodiversity have become direct threats to the financial stability and sustainability for companies, which are encouraged to announce the BioDD in their annual sustainability reports. Furthermore, a recent study dating back to World Economic Forum at Davos-Klosters from 19th to 23rd January, 2026 (<https://www.weforum.org/meetings/world-economic-forum-annual-meeting-2026/>) indicated that more than 50% of the international gross domestic product depends mainly and directly on natural resources, which does not make addressing biodiversity which is one of the most prominent potential future risks. In this context, there was a need to provide detailed information on environmental disclosure practices and their resulting impacts, including sectors in contact with the environment, such as oil, energy, and agriculture, which depend on the financial sector to finance their trade activities.

In the same vein, Environmental, Social, and Governance (ESG) practices have emerged as a critical framework for evaluating sustainable corporate performance. ESG encompasses non-financial criteria used to evaluate adherence to sustainability and risk-reduction standards, ultimately strengthening market positioning (Saleh et al., 2021). Furthermore, Velte (2022)'s study showed that environmental disclosure linked to sustainability standards has become a necessary factor in building trust between companies and stakeholders in European countries. The study also showed that governance plays a pivotal role in enhancing the reliability of these disclosures. In light of this, BioDD is no longer considered an optional practice, but has become an integral part of the corporate sustainability process especially after the emergence of the Global Reporting Initiative (GRI) and the Sustainability Accounting Standards Board (SASB), which encouraged companies to focus on environmental disclosures in the development related to biodiversity (Abe et al., 2025).

From an internal governance perspective, the risk management committee (RMC) represents an important mechanism for identifying and mitigating prospective risks. Today, the world, through international conferences and mutual visits by leaders of countries and major industrial companies, has become more interested in environmental risks such as climate change and biodiversity in light of the growing improvement in the operational environment of institutions and the increasing regulatory pressures associated with disclosing sustainability. Recent literature indicates that the existence of a specialized RMC whose members are experienced and independent contributes to enhancing the quality of oversight, improving the level of disclosure quality, and reducing opportunistic management behavior in both modern and developing systems (Akaro et al., 2024).

Cosma et al. (2022) conducted a study in European banks that highlighted weaknesses in climate change disclosure and board characteristics related to risk management and social accountability. Musa et al. (2025)'s study also found that RMCs play an important role in enhancing environmental disclosure and strengthening the responses of companies to climate risks and long-term sustainability. When addressing the agency theory, it highlights the importance of having effective oversight committees to reduce the information gap between management and investors, while the stakeholder theory confirms that companies seek to enhance their social legitimacy by disclosing more transparent environmental and social information (Kopnina et al., 2024). Studies represented the limited evidence that directly addressed risk management in promoting BioDD, specifically in developing economies and emerging markets (Kashif et al., 2026). Most of the previous studies were mainly concerned with the role of the board and the characteristics of management or the role of audit committees in environmental disclosure without addressing RMC (Jain & Kurian, 2025; Saleha et al., 2020; Saleh et al., 2025).

Prior literature has also primarily focused more on climate disclosure and carbon emissions (Di Vaio & Ali, 2025), while BioDD remains an emerging field that requires further study and analysis despite its growing importance in the international sustainability agenda (Kashif et al., 2026). Moreover, from an applied perspective, the relationship between the effectiveness of RMCs and the level of BioDD within the ESG framework remains unclear, particularly with regard to the contribution of these committees to improving the quality of disclosure, reducing environmental risks, and enhancing corporate sustainability. Hence the importance of the current study, as it seeks to bridge the gap via answering the study's importance by analyzing the role that the RMC can play in promoting BioDD within the ESG framework in developing countries—namely Iraq.

2. Literature Review

In sustainability requirements, BioDD involves how companies' activities affect ecosystems, natural resources, and living organisms on global markets (Şeker & İşgüven, 2025). Interest in this type of disclosure has increased as a result of the escalating environmental risks associated with biodiversity loss and climate change, as well as the growing pressure exerted by investors and international organizations on companies to enhance environmental transparency (Khandelwal & Khandelwal, 2025). The literature confirms that BioDD is an important tool for improving environmental accountability and reducing information asymmetries between management and stakeholders, as well as its role in supporting investment decisions and enhancing institutional reputation. In this context, Roberts et al. (2021) showed that companies with high BioDD enjoy higher levels of institutional legitimacy and market trust, as a result of their increasing response to global environmental pressures. According to Adler et al. (2018), most companies tended to focus on symbolic disclosure instead of substantive disclosure, indicating a disconnect between environmental discourse and actual practices.

Moreover, a study via Di Tommaso et al. (2025) indicated that companies' BioDD is directly linked to their ability to manage environmental risks, operational efficiency, and investor confidence in the long term, and that companies facing regulatory or societal pressures usually tend to expand disclosure of environmental biodiversity, other than climate disclosure or carbon emissions, to maintain their legitimacy in society (Zaid & Issa, 2024). Another study suggests that the quality of BioDD in non-financial reports is affected by many internal factors, such as governance structure, the effectiveness of the board of directors, and the independence of RMC (Khatib & Sulimany, 2026). For example, Pokharel (2025) argued that companies with independent RMCs are more experienced in dealing with complex risks and improving the quality of internal control systems. The study also indicated that the size of the company increases the likelihood of establishing specialized RMCs, which contributes to enhancing BioDD practices.

Despite this relative expansion in the literature, the majority of studies have focused on advanced economies particularly in Europe, while developing environments and emerging markets still suffer from a significant lack of applied studies related to this field despite its utmost importance. In addition, most previous studies have addressed the determinants of environmental disclosure in general, failing to examine in depth the role of RMCs in promoting BioDD under ESG requirements (Musa et al., 2025).

3. Hypotheses Development

3.1 Bank Size, Leverage, and Biodiversity Disclosure

Previous literature addressed the agency, legitimacy as well as stakeholders' theories have indicated that large companies are more vulnerable to societal, organizational, and stakeholder oversight compared to small companies, which prompts them to expand the scope of transparency in environmental, social and non-financial disclosure with the aim of preserving their image in the markets. In this context, several studies have shown that company size is one of the most important positive determinants of environmental disclosure and ESG disclosures (Bassen et al., 2025; Khalaf et al., 2026a). The authors found that large organizations tend to provide more quality disclosures about biodiversity than small businesses, because they have greater resources and higher regulatory capabilities that enable them to prepare more detailed sustainability reports. Evana et al. (2023)'s study also confirmed that company size represents an important explanatory variable for disclosing biodiversity and reducing agency costs or information asymmetry. Accordingly, it can be expected that company size is positively related to risk management disclosure and environmental governance.

Financial leverage represents one of the fundamental factors that affect the nature of institutional disclosure, as it reflects the extent of the company's dependence on debt financing, and the associated pressures exerted by lenders and creditors to enhance transparency and reduce risks. According to agency theory, companies with high financial leverage are more vulnerable to the problem of conflicts of interest between management and lenders, which prompts them to increase the level of disclosure to reduce agency costs and enhance lenders' confidence. Some studies have found a negative relationship between leverage and environmental disclosure in the European firms, as highly indebted companies tend to expand non-financial disclosure in order to reassure investors and lenders about their ability to manage environmental and sustainability risks (Bassen et al., 2025; Dănilă, 2025). In the same context, Noor & Sari (2026) study on the quality of ESG disclosure indicated that financial leverage directly affects disclosure and sustainability policies, according to which companies aim to adopt compliance standards and commit to transparency to improve their image in the face of the trust of stakeholders (Dănilă, 2025).

Other studies nevertheless have reached different results, indicating that institutions with high financial leverage may reduce environmental disclosure due to high compliance costs and financial pressures (Malik & Kashiramka, 2025). A study by Sharma et al. (2020) has shown that leverage can negatively impact the level of voluntary disclosure, especially in companies facing high financing expectations. This difference in results reflects the lack of conclusive consensus in the literature on the nature of the impact of size and leverage on environmental

disclosure and biodiversity, which justifies the need for further applied studies, especially in emerging markets and environments where environmental disclosures are still in the primitive stages of development. Therefore, the first hypothesis can be formulated as follows:

H1: *There is a statistically significant relationship between firm size and leverage with the level of BioDD.*

3.2 Risk Management Committee and Biodiversity Disclosure

RMC has become one of the most prominent modern governance mechanisms, especially after the growing environmental, climate and digital risks facing companies (Khalaf et al., 2026a). With the shift towards disclosing natural hazards, these committees have expanded to include oversight of sustainability issues and environmental risks associated with biodiversity. Recent literature indicates that the existence of effective RMCs contributes to improving the quality of non-financial environmental disclosure by strengthening control procedures and reducing information asymmetry in the long term upon the investors' expectations (Toukabri & Toukabri, 2025). For example, Hambali & Adhariani (2024) have investigated to more than 37 countries demonstrated the positive role of committees specializing in sustainability and risk management in disclosing biodiversity in environmentally sensitive industrial companies. Abe et al. (2025) also showed that companies with strong governance mechanisms are better able to provide higher-quality disclosures about biodiversity and environmental risk management and practice sustainability accounting. However, the study found a negative significant relationship between Sustainability Accounting Standards (SAP) and RMC. Moreover, there is no impact between the independence of the committee and SAP. Another evidence has shown that companies have sought to comply with international standards such as TNFD in light of growing global interest in nature and BioDDs as a key component of recent non-financial ESG reporting (Singhania & Gupta, 2024).

Although the literature on climate disclosure has expanded considerably in recent years comparatively insufficient attention has been devoted to BioDD and ESG governance (Eluyela et al., 2026). Most of these studies have addressed these variables separately, without integrating the impact of RMC, ESG governance and corporate financial characteristics within a unified explanatory model. The majority of recent studies have focused on advanced economies and European and American markets, while applied evidence related to emerging markets remains very limited. The shared relationship between RMC, ESG, company size, and leverage has also not received sufficient attention in recent literature, especially after the emergence of TNFD initiatives and nature-related disclosures. Based on this, the second hypothesis can be formulated as follows:

H2: *There is a significant positive relationship between the effectiveness of RMC, ESG, firm size, leverage, and level of BioDD.*

3.3 The Moderating Role of Environmental, Social, and Governance in the Relationship Between Risk Management Committee and Biodiversity

Recent literature suggests that the positive impact of RMC on environmental BioDD is significantly amplified in the presence of robust ESG systems that have a positive impact. Companies that adopt strong ESG practices often provide an institutional environment that supports the effectiveness of oversight committees in order to maintain their image in the face of stakeholders, by enhancing transparency and independence in disclosure and paying attention to long-term environmental risks (Qiu et al., 2026; Senanayake et al., 2024). Tao (2026) also documented that biodiversity has a positive impact on enhancing investor's confidence in Chinese companies as a result of the limited role played by the monitoring and RMC oversight is integrated under the ESG framework.

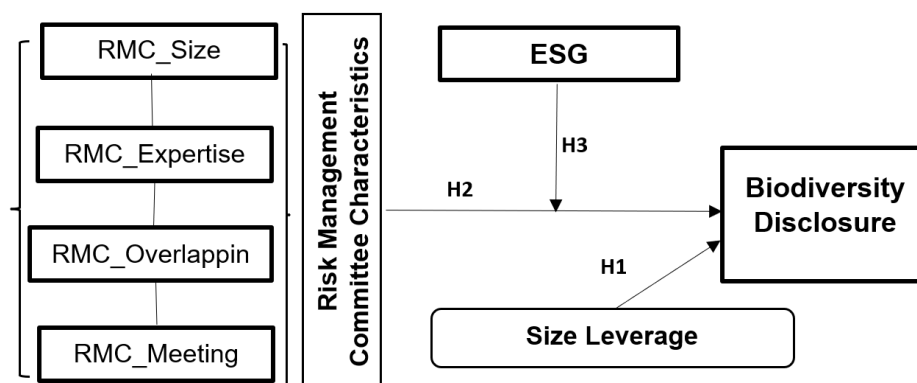


Figure 1. The hypothetical framework of the study

Note: RMC = Risk management committee; ESG = Environmental, Social, and Governance.

In developed countries, we find that Martini et al. (2026) have concluded that the experience of the Board of Directors and the RM significantly moderates the positive relationship between biodiversity and financial performance limiting the risk related to greenwashing in European companies across various sectors (Aman et al., 2026). On the contrary, Pokharel (2025) pointed out that economic institutions that participate in reducing the impact of biodiversity have a higher level of environmental performance based on the slight mediating role of ESG. ESG also contributes to reducing environmental costs in developed countries like New Zealand. Despite the prior evidence that focus on RMC and BioDD in developed countries, the literature lacks an ESG study to determine the nature of this effect in developing countries. This is what the current study seeks to address by formulating the following hypothesis:

H₃: ESG moderates the relationship between the RMC and BioDD level.

The main direction of the relationship between independent, dependent, moderating, control variables, as well as study hypotheses, can be seen in Figure 1.

4. Methodology

4.1. Sample and Data Collection

The study concentrates on the banks listed on the Iraqi Stock Exchange for the period from 2022 to 2025. The systematically collected data were obtained through the analysis of financial and sustainability reports published by the banks in the study sample. The listed banks were selected for their commitment to strict and transparent disclosure standards, as well as the availability of information on the exchange's official websites. Therefore, the number of Iraqi banks listed until 2025 is 49 banks. A final sample of 30 banks was selected, representing 61% of the total number of listed banks, after excluding banks whose data was not available due to outages or failure to publish financial reports. To conduct statistical analysis and achieve the study's objectives, a total of 120 observations were collected from the selected banks over the study period. This sample is sufficient and representative, as it represents the largest component of the Iraqi banking sector in terms of capital and turnover. This ensures the transparency and consistency of financial data, providing a solid basis for empirical analysis during a critical regulatory transition period (2022–2025).

4.2. Variables Measurement

4.2.1 Risk management committee

RMC was measured by four main dimensions, committee size based on number of members (Javaid et al., 2022). The committee's experience was compared to the number of members who had expertise in financial management and risk management among the total members (Almulhim et al., 2024). Meetings are frequently measured by the number of meetings held annually (Musa et al., 2025).

Hence, the RMC Index was built based on four main dimensions that represent the characteristics of the committee, namely: RMC_Size, RMC_Expertise, RMC_Overlapping, and RMC_Meeting. Each dimension was converted into a standard value with the aim of reducing the variance between the different measures. The dimensions were then given equal weights based on the Khalaf et al., (2026b) approach, which adopted the composite indicators approach. After that, the overall indicator was calculated through the arithmetic mean of the four dimensions so that the higher value reflects a higher level of effectiveness of the RMC.

4.2.2 Biodiversity disclosure

BioDD was assessed using a disclosure index consisting of 51 items divided into four categories (see Table A1), as referenced from previous literature (Alqubaysi & Musa, 2024; Elsayed, 2023; Skouloudis et al., 2019). Each BioDD item was assessed using a four-point weighted scale ranging from 0 to 3. A BioDD index value of 0 indicates the absence of biodiversity-related disclosure, while BioDD index value of 1, 2 and 3 reflects increasing levels of depth, accuracy and verifiability of disclosure. Unlike binary disclosure indicators, this approach takes into account the presence and quality of biodiversity-related information (Kopnina et al., 2024).

To ensure the applicability of this generic index to the financial context, industry-specific refinements were applied. Specifically, (a) all 51 items were retained as they conceptually align with the indirect environmental impacts of banking operations; however, items regarding operational 'biodiversity risk management' were adapted to reflect credit risk criteria, underwriting policies, and financed emissions. (b) Furthermore, to prevent scoring distortion, items deemed entirely outside a bank's scope were treated as 'not applicable' and excluded from that specific bank's denominator, rather than being penalized as 'non-disclosure' (zero score). (c) This adaptation ensures the index reflects the portfolio-driven nature of banking biodiversity exposure.

The final BioDD index value was calculated for each fiscal year as the weighted average of all disclosure items included in the BioDD index. Accordingly, the BioDD index represents a measure of BioDD quality rather than a disclosure ratio, as it captures the average quality, depth, specificity, and comprehensiveness of biodiversity-

related information disclosed by banks across the 51 disclosure items. Each item was assessed using a scoring scale ranging from 0 to 3, where higher values indicate more detailed and comprehensive disclosure practices. Therefore, higher BioDD values reflect stronger biodiversity reporting quality and greater usefulness of disclosed information for decision-making purposes. The descriptive statistics of BioDD should consequently be interpreted within the original scoring framework (0–3) rather than as percentages or proportions. This measurement approach allows the index to capture variations in the quality and extent of BioDD among banks rather than merely identifying the presence or absence of disclosure (Elsayed et al., 2024).

To ensure the reliability of the coding process, a double review was conducted on a sample of annual reports by independent researchers, and the coding results were compared and the degree of consistency between reviewers was verified using the Cohen's kappa coefficient, which reached a value of 0.78, indicating good inter-coder agreement. Furthermore, a cross-review process was conducted to improve coding consistency and reduce subjective judgement during the content analysis process.

4.2.3 Environmental, Social, and Governance

The ESG index was developed by using a disclosure-based approach (see Table B1) that captures the ESG disclosure practices reported by the sampled banks. Specifically, the raw data were extracted from the Commercial Banks' Performance Scorecard issued by Central Bank of Iraq (2025). The index consists of (18) disclosure indicators distributed over three dimensions: environmental (6 items), social (6 items), and governance (6 items). In line with previous ESG disclosure studies (Gai et al., 2023; Widyawati, 2020), equal weighting was applied to all indicators to avoid self-prioritization between dimensions. Each disclosed item was assigned one (1) value, while non-disclosed items were assigned zero (0). The ESG index was calculated as the ratio of disclosed ESG items to the total number of applicable ESG items, as represented in Eq. (1).

$$\text{ESG Index} = \sum \text{Disclosure Items} / \text{Total ESG Items} \quad (1)$$

Therefore, the ESG index reflects the extent to which banks disclose ESG-related information and should not be interpreted as a comprehensive assessment of their sustainability performance. To enhance the reliability of the measurement process, a double review of a sample of annual reports was conducted by independent researchers, the results were compared and the degree of consistency between reviewers was verified using Cohen's Kappa coefficient, which was 0.82, indicating very good agreement between reviewers. In addition, a mutual review process was adopted to ensure classification accuracy and reduce personal bias during content analysis.

4.2.4 Control variables

Bank size is the natural logarithm of total assets, while leverage is the total liabilities divided by total assets (Khalaf et al., 2026c; Khalaf & Hussein, 2024).

4.3 Study Models

To test the hypotheses, the following regression models were developed using the characteristics of the RMC as an independent variable, BioDD as a dependent variable, environmental and social governance as a moderating variable, and bank characteristics as control variables such as bank size and leverage.

$$\text{Model 1: } \text{BioDD } i = \beta_0 + \beta_1 \text{ SIZE } i + \beta_2 \text{ LEV } i + \varepsilon \quad (2)$$

$$\text{Model 2: } \text{BioDD } i = \beta_0 + \beta_1 \text{RMC } i + \beta_2 \text{ESG } i + \beta_3 \text{SIZE } i + \beta_4 \text{LEV } i + \varepsilon \quad (3)$$

$$\text{Model 3: } \text{BioDD } i = \beta_0 + \beta_1 \text{RMC } i + \beta_2 \text{ESG } i + \beta_3 (\text{RMC} * \text{ESG } i) + \beta_4 \text{SIZE } i + \beta_5 \text{LEV } i + \varepsilon \quad (4)$$

5. Data Analysis

5.1 Descriptive Analysis

The data in Table 1 present the descriptive statistics for the study variables, revealing that the RMC Index had a composite mean of 2.370, and specifically the RMC-Size averaged 3.533 members with a large standard deviation of 1.060, indicating variation in committee size across sampled banks. While RMC-Expertise recorded an average of 0.331 with a relatively low standard deviation of 0.072, meaning that about 33% of committee members have specialized experience, which is a positive indicator of oversight efficiency. The average RMC-Overlapping was 0.598 with a deviation of 0.158, indicating that nearly half of the members are members of other committees, which may be associated with lower committee independence. RMC-Meeting also shows an average

of 5.016 and a deviation of 1.321, with committees meeting approximately five times per year.

BioDD was measured as the average quality index value of BioDD based on a weighted scale ranging from 0 to 3, where higher values indicate greater depth, specificity, and comprehensiveness of disclosed biodiversity-related information. The descriptive statistics show that BioDD averaged 0.593 with a standard deviation of 0.324, while disclosure index value ranged from 0.353 to 1.471. These values indicate that BioDD practices among the sampled banks remain relatively limited, reflecting a generally low level of disclosure quality. However, the variation in index value suggests that some banks demonstrate relatively stronger transparency and more comprehensive biodiversity-related reporting practices than others. Because BioDD is calculated as a weighted average quality index rather than a binary disclosure ratio, values exceeding 1.0 remain theoretically possible and indicate higher levels of disclosure quality and comprehensiveness.

In contrast, the average ESG disclosure score was 0.743, with a standard deviation of 0.152, while the scores ranged from 0.417 to 1.000 across the sampled banks. These results indicate that Iraqi listed banks generally exhibit a relatively high level of ESG-related disclosure compared with the theoretical scoring range of the index. However, the variation among banks suggests differences in the extent and comprehensiveness of ESG information reported in their annual and sustainability-related disclosures. Since the ESG index is constructed as a disclosure-based measure using equally weighted ESG indicators, the reported values should be interpreted as the level of ESG disclosure transparency rather than as a direct measure of the level of ESG disclosure transparency rather than as a direct measure of actual ESG performance or regulatory compliance.

Finally, the average size of banks was 2.530, yet the standard deviation was 0.405. While the leverage is 0.365 with a relatively low standard deviation rate of 0.158.

Table 1. Descriptive statistics

Variable	N	Mean	Median	Std. Dev.	Minimum	Maximum
RMC Index	120	2.370	2.166	0.466	1.417	3.450
RMC—Size	120	3.533	3.000	1.060	2.000	7.000
RMC—Expertise	120	0.331	0.333	0.072	0.143	0.500
RMC—Overlapping	120	0.598	0.666	0.158	0.333	1.000
RMC—Meeting	120	5.016	4.000	1.321	2.000	8.000
BioDD	120	0.593	0.411	0.324	0.353	1.471
ESG	120	0.743	0.750	0.152	0.417	1.000
SIZE	120	2.530	2.505	0.405	1.133	3.400
LEV	120	0.365	0.357	0.158	0.000	0.757

Note: RMC = Risk management committee; BioDD = Biodiversity disclosure; ESG = Environmental, Social, and Governance; SIZE = Firm size; LEV = Leverage.

5.2 Data Testing

Table 2 presents in-depth results of examining multicollinearity between study variables through Variance Inflation Factor (VIF) and tolerance values, as well as testing normal distribution (skewness and kurtosis). This analysis aims to identify and address any statistical issues that may affect the robustness and accuracy of regression models. In our results, all VIF values ranged between (1.169–4.389), which is well below the critical threshold of 10, while tolerance values ranged between (0.228–0.855). A VIF value greater than 5 and a tolerance value less than 0.2 indicate multicollinearity correlation as stated by Khalaf et al. (2026c). Finally, the moderation test shows that all skewness and kurtosis values were within the statistically acceptable limits specified in (Khalaf & Hussein, 2024), indicating that the data are close to a normal distribution. Furthermore, the assumptions of the regression model were met, allowing hypotheses to be tested with statistical confidence.

Table 2. Data testing

Variable	N	Normality Test		Multicollinearity Test	
		Skewness	Kurtosis	VIF	Tolerance
RMC index	120	0.511	-0.644	3.295	0.303
RMC—Size	120	0.544	1.936	4.389	0.228
RMC—Expertise	120	-0.260	1.712	2.157	0.464
RMC—Overlapping	120	0.061	0.460	2.341	0.427
RMC—Meeting	120	0.324	-0.618	3.137	0.319
BioDD	120	0.541	1.072	N/A	N/A
ESG	120	-0.615	-0.530	1.415	0.707
SIZE	120	-0.965	2.233	3.647	0.274
LEV	120	-0.143	0.721	1.169	0.855

Note: RMC = Risk management committee; BioDD = Biodiversity disclosure; ESG = Environmental, Social, and Governance; SIZE = Firm size; LEV = Leverage; VIF = Variance Inflation Factor; N/A = Not applicable.

5.3 Correlation Test

The correlation matrix in Table 3 provides preliminary evidence of relationships between variables, and the absence of high correlation coefficients indicates the absence of a multicollinearity problem, supporting the reliability of subsequent econometric models. The results indicate that the relationship between RMC and BioDD characteristics is positive and strongly significant ($R = 0.886$, $\text{Sig} = 0.000$), as the size of the committee and its meetings are among the characteristics that contributed significantly to this association, which means that banks which are characterized by stronger supervision of risk management tend to show high levels of BioDD. The results also showed significant positive correlations between ESG and both RMC and BioDD ($R = 0.473$, $\text{Sig} = 0.000$) and with BioDD ($R = 0.444$, $\text{Sig} = 0.000$), suggesting that banks with a greater environmental, social and governance orientation generally tend to pay attention to the characteristics of RMC and BioDD.

Table 3. Correlation analysis

Coefficient	1	2	3	4	5	6	7	8	9
(1) RMC	1								
(2) RMC_Size	0.729**	1							
(3) RMC_Expertise	-0.076	-0.491**	1						
(4) RMC_Overlapping	-0.326**	-0.428**	-0.106	1					
(5) RMC_Meeting	0.871**	0.305**	0.244**	-0.231*	1				
(6) BioDD	0.886**	0.847**	-0.148	-0.371**	0.625**	1			
(7) ESG	0.473**	0.301**	-0.097	-0.058	0.439**	0.444**	1		
(8) SIZE	0.826**	0.580**	-0.102	0.011	0.705**	0.720**	0.519**	1	
(9) LEV	0.154	0.253**	-0.256**	0.078	0.019	0.191*	0.269**	0.305**	1

Note: ** A strong significant relationship ($\text{Sig.} < 0.01$) and * significant relationship ($\text{Sig.} < 0.05$); RMC = Risk management committee; BioDD = Biodiversity disclosure; ESG = Environmental, Social, and Governance; SIZE = Firm size; LEV = Leverage.

5.4 Hypotheses Test

Table 4 presents the results of the study hypotheses in three stages using multiple hierarchical regression analysis. In the first stage, the effect of the control variables SIZE and LEV on BioDD was tested as shown in Model 1, and in the second stage, the RMC and ESG variables were added to examine the effect of the main variables on BioDD as shown in Model 2. In the third stage, the interaction variable $\text{RMC} \times \text{ESG}$ was added to explore the interaction effect on BioDD as shown by Model 3.

Table 4. Hierarchical regression analysis

Variables	Model 1 Controls	Model 2 Main Effects	Model 3 Interaction Effect
SIZE	0.729***	-0.086	0.008
LEV	-0.032	0.065	-0.100*
RMC	N/A	0.932**	-0.563*
ESG	N/A	0.030	-1.221***
$\text{RMC} \times \text{ESG}$	N/A	N/A	2.339***
Observations	120	120	120
Year Effect	Yes	Yes	Yes
R^2	0.519	0.790	0.825
Adjusted R^2	0.511	0.783	0.818
ΔR^2	0.519***	0.271***	0.035***
F-value	63.089***	108.316***	107.749***
F Change	63.089***	74.393***	22.915***
Sig. F Change	0.000	0.000	0.000

Note: Dependent variable: Biodiversity disclosure (BioDD). Reported coefficients are standardized beta coefficients. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$; RMC = Risk management committee; ESG = Environmental, Social, and Governance; SIZE = Firm size; LEV = Leverage; N/A = Not included in the corresponding model.

The results reveal that Model 1 is statistically significant ($F = 63.089$, $p < 0.01$), suggesting that the selected control variables jointly explain the differences in BioDD practices between banks. The explanatory power of the model is relatively high, with an R^2 value of 0.519, indicating that approximately 51.9% of the variance in BioDD can be explained by the control variables alone. Regarding individual effects, SIZE shows a positive and significant effect on BioDD ($B = 0.729$, $p < 0.01$). This result indicates that larger banks are more likely to offer large-scale BioDD. In contrast, LEV ($B = -0.032$, $p > 0.10$) does not appear and this result indicates that banks' capital structure does not appear to have a decisive role in shaping BioDD practices.

Model 2 expands the framework model by integrating the main variables RMC and ESG in addition to the control variables. The results show that the Model is statistically significant ($F = 108.316$, $p < 0.01$). Including

these main variables significantly improves the explanatory power of the model in terms of the increase in the value of R^2 from 0.519 to 0.790. The change in explanatory power ($\Delta R^2 = 0.271$, $p < 0.01$) indicates that the newly added variables RMC and ESG explained an additional 27.1% of the variance in BioDD beyond the effects of SIZE and LEV. Regarding individual effects, RMC exerts a positive and significant effect on BioDD ($B = 0.932$, $p < 0.05$), and conversely, ESG does not show a direct statistically significant effect on BioDD ($B = 0.030$, $p > 0.10$).

Finally, Model 3 introduces the interaction term $RMC \times ESG$ to examine the moderating effect of ESG disclosure. The results show that the model is statistically significant ($F = 107.749$, $p < 0.01$), with an increase in explanatory power from $R^2 = 0.790$ to 0.825 ($\Delta R^2 = 0.035$, $p < 0.01$). The interaction coefficient is positive and statistically significant ($\beta = 2.339$, $p < 0.01$), indicating that ESG disclosure strengthens the positive association between RMC and BioDD. The lower-order coefficients should be interpreted in the context of the interaction effect rather than independently, as the significant interaction term provides evidence supporting the moderating role of ESG disclosure.

6. Discussion

This section focuses on interpreting empirical findings and comparing them with existing literature, using agency theory, stakeholder theory, and legitimacy theory to analyze the findings, and emphasizing their practical importance to the banking sector in Iraq. The results were examined through a series of hypotheses, starting with bank characteristics such as SIZE and LEV, then RMC and ESG, and ending with the effect of ESG as an interactive variable ($RMC \times ESG$) on these relationships, demonstrating the theoretical and practical contributions of the study.

The results indicated that bank size is one of the factors that is positively associated with BioDD practices, confirming that larger banks are more inclined to increase the level of BioDD compared to small banks. This is consistent with the legitimacy theory, which assumes that large organizations are more vulnerable to societal, organizational, and media pressures, which prompts them to improve the level of transparency and disclosure to maintain their legitimacy and continued social acceptance (Abdullah et al., 2024). Conversely, financial leverage did not exert a significant impact on BioDD, which indicates that creditors and lenders in the environment under study do not view BioDD as an essential factor in assessing risks and making financing decisions. This finding is partly consistent with the logic of agency theory, which assumes that managers in institutions with high financial liabilities focus on aspects that reduce direct financial pressures rather than investing in BioDD practices (Amara et al., 2025; Jo & Harjoto, 2012).

The study further revealed a positive association between RMC and BioDD, reflecting the pivotal role played by internal governance mechanisms in enhancing transparency. Within the framework of agency theory, the characteristics of RMC contribute to reducing the information gap between management and stakeholders by strengthening oversight and supervision processes for emerging risks, including climate and environmental risks. Effective RMC characteristics may facilitate the integration of biodiversity considerations into enterprise risk management systems, which is associated with higher levels of BioDD. This finding is consistent with a study (Suwaid et al., 2026; Thi & Dai, 2025) which confirmed that stakeholder theory assumes that organizations seek to respond to the expectations of various stakeholders by adopting more transparent disclosure practices regarding biodiversity issues. In contrast, the results showed that the direct impact of ESG on BioDD was not significant in Model 2. This indicates that ESG indicators are not directly reflected in BioDD because they encompass broad and multidimensional aspects, including social and governance dimensions alongside environmental ones. Consequently, the specific focus on BioDD is often neither clear nor direct within these composite indicators (Xin et al., 2025). This also reflects a discrepancy between adopting ESG standards at the formal level and actually integrating them into BioDD strategies, this finding is consistent with some recent literature that has confirmed that BioDD remains a relatively emerging aspect of ESG reporting, particularly in emerging countries and the banking sector (Xin et al., 2025; Yang & Wu, 2025).

Finally, the introduction of the interaction variable ($RMC \times ESG$), in model 3 showed an important shift, it should be noted that the negative coefficients of the direct coefficients of both RMC and ESG, after introducing the interaction variable ($RMC \times ESG$), should not be interpreted as final direct relationships, because the presence of the interaction variable in the model makes these coefficients represent conditional effects, depending on the value of the other interaction variable. Moreover, the main focus in interpreting Model 3 should be on the significance and direction of the interaction variable itself and not on individual transactions. As it becomes clear that this interaction has a significant impact on BioDD, and thus the efficiency of RMC becomes greater in banks that have high levels of ESG disclosure, indicating that ESG significantly moderates the relationship between RMC and BioDD, such that the association between RMC and BioDD becomes stronger at higher levels of ESG disclosure, and this findings confirms the validity of the stakeholder theory (Elgharbawy & Aladwey, 2025; Khalaf et al., 2026c) which assumes that banks with a high ESG disclosure orientation are more sensitive to the expectations of investors, customers and regulators regarding biodiversity issues making RMC more effective in

directing management towards BioDD. Within the framework of legitimacy theory (Abdullah et al., 2024), banks seek to adopt more ESG standards in order to improve their corporate image and maintain their societal legitimacy by improving the level of BioDD. In general, BioDD in the banking sector does not depend only on the presence of an effective RMC, but it is also affected by the extent to which the characteristics of the RMC are integrated with the institutional trends related to ESG.

Despite growing global interest in sustainability and corporate ESG reporting, the Iraqi banking sector remains at a relatively early stage of institutional integration of these practices. However, in recent years, notable efforts have emerged to enhance governance and transparency within the banking sector through regulatory reforms led by the Central Bank of Iraq and the increasing adoption of governance guidelines by listed banks. Furthermore, the expansion of digital banking, growing interest in sustainable finance, and gradual alignment with international reporting practices have encouraged banks to improve the quality of non-financial disclosures. However, biodiversity reporting and ESG disclosures remain largely voluntary, with limited regulatory requirements and significant variation in reporting practices across institutions. This unequal level of implementation may explain why the relationships identified in the current study differ from those reported in more developed markets, where sustainability reporting frameworks are more mature, ESG integration is more institutionalized, and disclosure requirements are generally subject to stronger regulatory oversight. Similar observations have been highlighted in recent studies examining sustainability reporting practices in emerging economies, which indicates that mature institutions play an important role in shaping the effectiveness of governance mechanisms and disclosures related to sustainability (da Cunha et al., 2025; Thi & Dai, 2025).

7. Conclusion

This study examined the association between RMC characteristics and BioDD among banks listed on the Iraq Stock Exchange, and investigated the moderating role of ESG disclosure practices in this relationship. The empirical results indicate that RMC characteristics are positively associated with BioDD, and that ESG disclosure significantly moderates this association—specifically, the positive RMC–BioDD relationship is stronger at higher levels of ESG disclosure. These findings suggest that in the Iraqi banking context, governance mechanisms and sustainability disclosure practices are interrelated in ways that may support more transparent biodiversity-related reporting.

Although the banking sector does not generate direct environmental impacts comparable to those of heavy industries, banks are indirectly exposed to biodiversity-related risks through their lending portfolios and project financing activities. Financing resource-intensive projects and extractive industries may expose banks to elevated environmental and regulatory risks. The growing global emphasis on sustainable finance therefore provides an additional impetus for banks to strengthen their environmental and ESG-related disclosures, including those pertaining to biodiversity.

The findings carry several practical implications for regulators, bank management, and investors. From a regulatory standpoint, the results highlight the value of developing BioDD-specific disclosure frameworks within the banking sector, extending beyond traditional financial indicators to include biodiversity-related requirements in periodic and sustainability reports. At the administrative level, expanding RMC responsibilities to encompass environmental, climate, and biodiversity risks may further strengthen banks' risk oversight capabilities. For investors and stakeholders, BioDD may serve as one indicator of a bank's governance quality and its commitment to sustainability practices.

This study contributes to the BioDD literature in three ways. First, it provides empirical evidence from Iraq, an emerging-market context where such research remains scarce. Second, it integrates RMC characteristics and ESG disclosure practices within a unified analytical framework, an approach largely absent in prior studies. Third, it offers empirical support for agency, stakeholder, and legitimacy theories as complementary lenses for understanding BioDD in a banking context. Future research could extend this work by examining other sectors and countries, disaggregating ESG into its ESG sub-dimensions, incorporating additional governance variables (e.g., board diversity, digital transformation), and exploring the relationship between BioDD and firm value.

The findings should be interpreted within the institutional context of the Iraqi banking sector and the specific measurement framework adopted in this study. The reported relationships reflect statistical associations identified within this research setting and should not be interpreted as universal causal effects beyond this context.

Author Contributions

Conceptualization, A.J.K. and S.R.J.; methodology, A.F.S. and A.M.Kh.; software, A.J.K., A.M.Kh., and M.I.A.; validation, A.S.M. and A.S.A.; formal analysis, D.A.A.; investigation, A.J.K.; resources, A.F.S.; data curation, S.R.J., A.M.Kh., M.I.A., D.A.A., and A.M.Ka.; writing—review and editing, A.F.S.; visualization, A.J.K.; supervision, A.J.K.; project administration, A.F.S.; funding acquisition, S.R.J., A.M.Kh., M.I.A., D.A.A., and A.M.Ka. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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Appendix

Table A1. Structure of the biodiversity disclosure (BioDD) index

Main Dimension	Sub-Dimension Code	Disclosure Category	Description of Disclosure Content	Number of Items
Dimension 1: Current and Historical Biodiversity Actions	BioDD1	Biodiversity Policies and Conservation Initiatives	Disclosure of biodiversity policies, conservation commitments, biodiversity protection programs, and environmental stewardship initiatives.	5
	BioDD2	Stakeholder Engagement and Biodiversity Awareness	Disclosure of partnerships, community engagement activities, awareness campaigns, collaborations with environmental organizations, and stakeholder involvement in biodiversity protection.	4
	BioDD3	Habitat Restoration and Ecosystem Protection	Disclosure of habitat restoration projects, ecosystem conservation measures, biodiversity monitoring activities, and species protection programs.	5
	BioDD4	Biodiversity Performance and Operational Management	Disclosure of biodiversity-related performance outcomes, operational biodiversity management practices, environmental impact assessments, and biodiversity monitoring results.	9
Subtotal (Dimension 1)				23
Dimension 2: Future Biodiversity Commitments and Risk Mitigation	BioDD5	Strategic Biodiversity Objectives and Future Commitments	Disclosure of future biodiversity goals, strategic plans, biodiversity targets, and long-term conservation commitments.	3
	BioDD6	Biodiversity Risk Management and Preventive Measures	Disclosure of biodiversity risk assessments, mitigation strategies, preventive actions, and biodiversity-related contingency planning.	4
	Subtotal (Dimension 2)			7
Dimension 3: Financial Biodiversity Disclosure	BioDD7	Biodiversity Investments and Resource Allocation	Disclosure of expenditures, investments, budgets, and financial resources allocated to biodiversity conservation and environmental protection activities.	2
	BioDD8	Biodiversity Liabilities and Legal Exposure	Disclosure of environmental provisions, biodiversity-related penalties, legal claims, litigation risks, and environmental liabilities.	3
	Subtotal (Dimension 3)			5
Dimension 4: Biodiversity Governance and Accountability Framework	BioDD9	Biodiversity Governance Integration	Disclosure of the integration of biodiversity considerations into governance structures, board oversight, and managerial decision-making processes.	4
	BioDD10	Compliance with Biodiversity Standards and Frameworks	Disclosure of alignment with sustainability standards, biodiversity reporting frameworks, environmental regulations, and international best practices.	4
	BioDD11	Biodiversity Performance Measurement Systems	Disclosure of biodiversity indicators, monitoring mechanisms, performance evaluation systems, and biodiversity-related key performance indicators (KPIs).	4
	BioDD12	Quantitative and Visual Biodiversity Reporting	Disclosure of biodiversity metrics, statistical information, maps, charts, species data, and other visual reporting tools used to communicate biodiversity performance.	4
Subtotal (Dimension 4)				16
Total Biodiversity Disclosure Items (BioDD) Index				51

Note: The biodiversity disclosure (BioDD) index consists of 51 disclosure items distributed across four major dimensions. Each item is evaluated using a weighted four-point scale ranging from 0 to 3, where 0 indicates no disclosure and 3 indicates comprehensive, specific, and verifiable disclosure. The final BioDD index value is calculated as the weighted average of all disclosure items, reflecting the overall quality and comprehensiveness of biodiversity disclosure practices.

Table B1. Environmental, Social, and Governance (ESG) disclosure index

Dimension	Code	Measurement Indicator	Number of Items
Environmental (E)	E1	Environmental policy and sustainability commitments	6
	E2	Energy efficiency and resource conservation initiatives	

	E3	Climate change and emissions disclosures	
	E4	Biodiversity and ecosystem protection activities	
	E5	Environmental compliance and risk management	
	E6	Green financing and sustainable investment initiatives	
	S1	Employee welfare and occupational health policies	
	S2	Training, education, and human capital development	
Social (S)	S3	Diversity, equality, and inclusion practices	6
	S4	Community engagement and social responsibility activities	
	S5	Customer protection and service quality initiatives	
	S6	Stakeholder engagement mechanisms	
	G1	Board independence and governance structure	
	G2	Risk management and internal control systems	
Governance (G)	G3	Audit committee effectiveness	6
	G4	Ethical conduct and anti-corruption policies	
	G5	Transparency and disclosure practices	
	G6	Shareholder rights and governance accountability	
Total Index	ESG		18
Scoring Method:			
1 = Item disclosed			
0 = Item not disclosed			
ESG Index = Total Disclosed Items ÷ Total Applicable Items			